

Interference Measurement Between 3.5 GHz 5G System And Radar

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Abstract— Interference test was performed between the 3.5 GHz 5G system assigned recently and the radar operating in adjacent band. For interference test in the field, a 3.5 GHz 5G system was installed and the interference effect was analyzed with the data rate of the 5G system when the radar transmitted three transmission modes individually. From a results of the test, we found that uplink data rates of 5G system at UE RSRP=-100 dBm were almost not affected by interference, although the radar received powers were as high as -52.9~74.6 dBm/MHz. This was analyzed to be due to characteristics of radar signal such as narrow antenna beam width and short pulse width, and error recovery ability of 5G mobile system.

Keywords—Interference measurement, 5G, Radar

I. INTRODUCTION

5G system is a next generation mobile communication system that data traffic capacity is about 1000 times faster and speed is about 200 times faster than the 4G LTE system, and can provide services such as AR, VR, real time online game etc. Many countries have been competing to provide 5G mobile communication services first. However, when assigning new frequency band for the 5G system, it is important that interference does not occur between the 5G system and the system operating already in adjacent band.

In Europe, there was decided to allocate 5G system to 3.4~3.8 GHz band. ECC studied interference between 5G system using active antenna system and radiolocation, airborne radar, and military radar operating in adjacent band[1]. CEPT proposed that the transmitter out-of-block restricted baseline power limit should be further strengthened to protect radars operating in adjacent bands[2].

Korea is striving to provide the world's first 5G mobile service. For this purpose, new 3.5 GHz band and 28 GHz band for 5G have been assigned to mobile operators. The Original plan was to assign a 300 MHz bandwidth in the 3.5 GHz band, but it was reduced to 280 MHz bandwidth in order to avoid interference from various types of radar operating in adjacent band[3].

In this paper, we have installed a 3.5 GHz band 5G mobile communication system in the field, and analyzed interference effect between 5G system and radar operating in adjacent band with the transmission data rate of 5G system.

II. INTERFERENCE MEASUREMENT SYSTEM

A. 5G System

In order to analyze the interference effect of the radar signal on the 5G mobile communication system, a 5G BTS satisfying 3GPP standard[4] was installed and 5G UE[5] was used. The active antenna structure of the 5G BTS was 64T64R and the maximum output power was 73 dBm. The active antenna structure of the 5G UE was 8T8R and the maximum output power was 29 dBm. The channel bandwidth was 100 MHz and SCS=60 kHz. The Figure 1 shows the 100 MHz bandwidth transmission spectrum of 5G BTS used in this test.

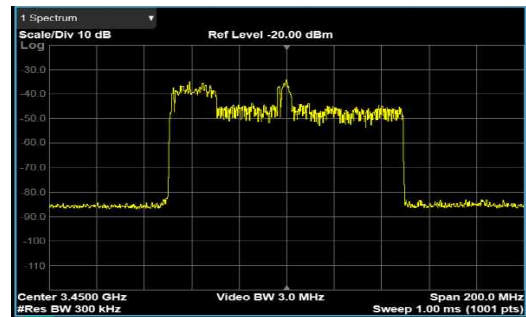


FIGURE 1. TRANSMISSION SPECTRUM OF 5G BTS 100 MHz BANDWIDTH

B. Radar Measurement System

Since the radar signal to be measured is a pulse-shaped signal and transmit at an irregular time period, RTSA(Real Time Spectrum Analyzer) which can measure all incoming signals in time without missing was used[6]. When we used a directional antenna, the minimum measurable level of the radar measurement system is about -119 dBm/MHz.

III. INTERFERENCE TEST SCENARIO

The interference test scenario measures the transmission rate of the 5G system at UE RSRP(Reference Signals Received Power) = -100 dBm when the radar in the sea is transmitting in various transmission modes as shown in Figure 2. There are three transmission modes of the radar: (1) High power mode, (2) Low power mode, and (3) Proximity mode. Proximity mode is when the radar transmits near the shore.

The radar transmission power in the High power mode is larger than in the Low power mode or Proximity mode, but the transmission frequency channel of radar and the distance between the radar and the measuring points on the shore different for each mode. So it should be analyzed based on radar received power measured on the coast rather than radar transmission power. Radar received power imply the received power at the 5G receiver for the signal transmitted by radar. The radar received power refers to the power measured by the radar measurement system at the shore for the signal transmitted by the radar.

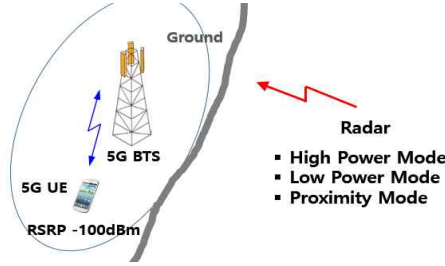


FIGURE 2. INTERFERENCE TEST SCENARIO

IV. INTERFERENCE TEST RESULT

A. In case of the radar transmitting weakly

The Table I and II were the measurement result of the uplink and downlink data rate of 5G UE at RSRP=-100 dBm and the radar received power when there was a weak radar signal. This was tested in advance to compare with the test results when the radar transmitted in three transmission modes. The radar received power measured the received power in the radar frequency channel closest to the 5G band. The radar received powers were about -97~-100 dBm/MHz in both uplink and downlink. 5G uplink data rate and downlink data rates were measured about 14~19.6 Mbps, 74~119 Mbps, respectively.

TABLE I. UPLINK MEASUREMENT DATA WITH WEAK RADAR

No	Radar Received Power(dBm/MHz)		Data Rate (Mbps)	MCS	RSRP (dB)
	Max	Ave			
Test 1	-99.6	-100.1	15.8	12.4	-99.6
Test 2	-94.6	-97.9	16.7	13.2	-100.0
Test 3	-91.2	-97.1	19.6	14.7	-99.6
Test 4	-92.2	-99.3	14.0	11.1	-97.9
Test 5	-96.4	-99.7	14.0	11.2	-98.0
Test 6	-96.6	-99.5	14.5	11.5	-98.0

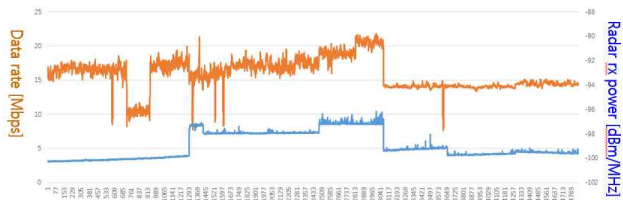


FIGURE 3. THE RADAR RX POWER AND DATA RATE OF ALL UPLINK TESTS

Figure 3 and 4 show the instantaneous radar receive power and instantaneous data rate of all tests together. We can see that the measured instantaneous values are slightly different between each test. This indicates that the 5G system performance or test environment may changes a little between tests.

TABLE II. DOWNLINK MEASUREMENT DATA WITH WEAK RADAR

No	Radar Received Power(dBm/MHz)		Data Rate (Mbps)	MCS	RSRP (dB)	SINR (dB)
	Max	Ave				
Test 1	-99.6	-100.1	74.1	10.3	-99.8	20.2
Test 2	-94.6	-97.9	101.6	12.0	-99.8	20.3
Test 3	-87.0	-97.3	80.0	10.7	-100.0	20.3
Test 4	-93.4	-99.2	102.1	12.0	-98.4	20.8
Test 5	-94.7	-99.5	119.6	13.0	-97.4	21.0
Test 6	-94.7	-99.7	115.4	12.9	-97.7	20.9



FIGURE 4. THE RADAR RX POWER AND DATA RATE OF ALL DOWNLINK TESTS

B. In case of the radar transmitting in three modes

The Table III was the measurement result of the uplink data rates of 5G UE at RSRP= -100 dBm when the radar was transmitting in three transmission modes. The radar leakage power was the measurement of radar out-of-band power in 3.4~3.5 GHz band which was part of the 5G band. In High power mode, Low power mode and Proximity mode, the radar received powers were measured about -73 dBm/MHz, -72 dBm/MHz, -53 dBm/MHz, respectively and radar leakage powers were measured about -115 dBm/MHz, -114 dBm/MHz, -95 dBm/MHz, respectively. The uplink data rates of the 5G system for each mode were measured about 14.6 Mbps, 16.2 Mbps, 14.9 Mbps, respectively.

TABLE III. UPLINK MEASUREMENT DATA IN THREE RADAR MODES

Radar Tx Mode	No	Radar Leakage Power(dBm/MHz)		Radar Received Power(dBm/MHz)		Data Rate (Mbps)	MCS	RSRP (dB)
		Max	Ave	Max	Ave			
High Power	Test 1	-95.4	-114.2	-53.4	-72.2	15.0	11.8	-99.0
	Test 2	-101.5	-116.6	-59.5	-74.6	14.1	11.5	-99.0
Low Power	Test 3	-98.0	-112.8	-56.0	-70.8	18.5	14.1	-98.7
	Test 4	-99.9	-115.3	-57.9	-73.3	13.8	11.0	-99.8
Proximity	Test 5	-94.7	-94.9	-52.7	-52.9	14.9	12.0	-98.6

The radar received spectrums measured on the coast for each mode are Figure 5, 7, and 9. These were measured with RTSA and showed the cumulative value measured for 10 minutes.

Figures 6, 8, and 10 show instantaneous data of all the tests measured in each mode.

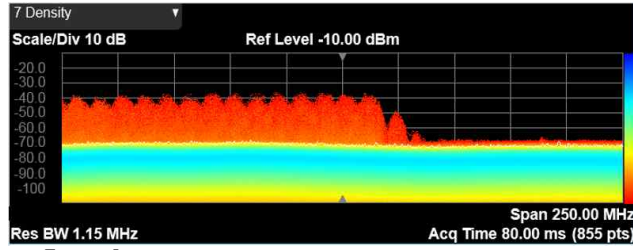


FIGURE 5. RADAR RECEIVED POWER SPECTRUM IN HIGH POWER MODE

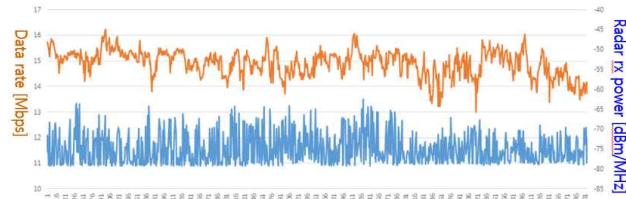


FIGURE 6. RADAR RX POWER AND UPLINK DATA RATE IN HIGH POWER MODE

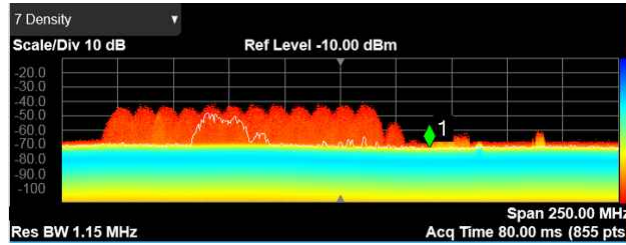


FIGURE 7. RADAR RECEIVED POWER SPECTRUM IN LOW POWER MODE

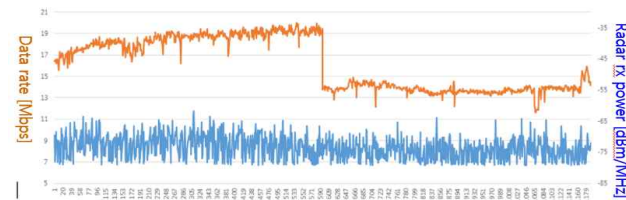


FIGURE 8. RADAR RX POWER AND UPLINK DATA RATE IN LOW POWER MODE

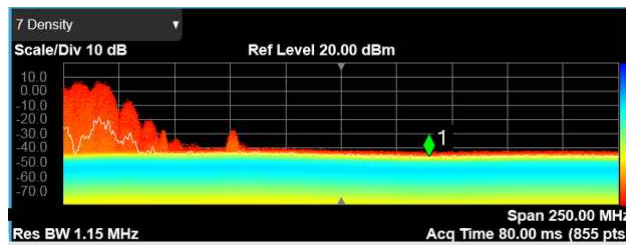


FIGURE 9. RADAR RECEIVED POWER SPECTRUM IN PROXIMITY MODE

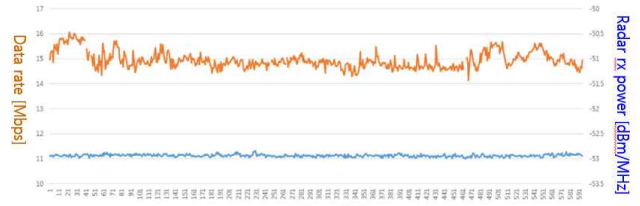


FIGURE 10. RADAR RX POWER AND UPLINK DATA RATE IN PROXIMITY MODE

The radar received powers were measured on the coast as -97.1~-100.1 dBm/MHz for weak case, and -52.9~74.6 dBm/MHz for three transmission modes case. Although the level difference of radar received power was large, the uplink data rates at RSRP=-100dBm were measured similar to each other as 14~19.6 Mbps for weak and 13.8~18.5 Mbps for three transmission modes case. It can be seen that the radar system used in this test did not interfere with the 5G system.

V. CONCLUSION

In this paper, we analyzed the interference effect between 5G mobile communication system in the 3.5 GHz and the radar operating in adjacent band. For the test, we constructed a 3.5 GHz 5G system and analyzed the interference effect with the data rate of the 5G system when the radar transmitted three transmission modes individually. From a result of measurement at the UE RSRP=-100 dBm, the data rate was maintained constantly even though a relatively large radar signal was received. This was analyzed to be due to characteristics of radar signal such as narrow antenna beam width and short pulse width, and error recovery ability of 5G mobile system.

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